Unit: 6 Unit 8: Quadratics **Name:** ______________

Topic: Definition of Quadratics and Standard Form **Date:** __________

Advanced (Habanero)

Q1. What is the standard form of a quadratic function?

(A) $ax^2 + bx + c$

(B) $ax + bx^2 + c$

(C) $ax^2 - bx + c$

(D) $ax^3 + bx^2 + c$

(E) $ax^2 + b$

Q2. Which of the following is a quadratic function in standard form?

(A) $x^2 + 3x - 2$

(B) $x^3 - 2x^2 + x$

(C) $x^2 - 4$

(D) $x + 2$

(E) $x^4 + x^2$

Q3. What is the value of a in the quadratic function $2x^2+3x-1$?

(A) 1

(B) 2

(C) 3

(D) -1

(E) -2

Q4. Which of the following quadratic functions has a negative leading coefficient?

(A) $2x^2 + 3x - 1$

(B) $-2x^2 + 3x - 1$

(C) $x^2 - 2x + 1$

(D) $x^2 + 2x - 1$

(E) $-x^2 - 2x + 1$

Q5. What is the axis of symmetry of the quadratic function $x^2 + 2x + 1$?

(A) $x = -1$

(B) $x = 1$

(C) $x = 0$

(D) $x = 2$

(E) $x = -2$

Q6. Which of the following is a key feature of the quadratic function $x^2 + 2x + 1$?

(A) A maximum value

(B) A minimum value

(C) A zero at $x = -1$

(D) A vertical asymptote

(E) A horizontal asymptote

Q7. What is the vertex of the quadratic function $x^2 + 2x + 1$?

(A) $(-1, 0)$

(B) $(-1, -1)$

(C) $(-2, 0)$

(D) $(-1, 1)$

(E) $(-1, 2)$

Q8. Which of the following quadratic functions has a vertex at $(0, 0)$?

(A) $x^2 + 2x + 1$

(B) $x^2 - 2x - 1$

(C) x^2

(D) $x^2 + 1$

(E) $x^2 - 1$

Open Ended Questions (FRQ)

Q9. Describe the key features of the quadratic function $x^2 + 4x + 4$. Provide the axis of symmetry, vertex, and any zeros.

Q10. Write a quadratic function in standard form with a vertex at $(-2, 3)$ and a leading coefficient of 2. Explain your reasoning.

Chef's Tips

- Ensure students understand the definition of a quadratic function in standard form.
- Emphasize the importance of identifying key features such as axis of symmetry, vertex, and zeros.
- Encourage students to use graphing calculators or software to visualize and explore quadratic functions.